

Estimated Annual Agricultural Pesticide Use Madison County, Idaho 2014-2018

24 July 2026

Summary

Madison County ranks **#3 of Idaho's 44 counties** by total estimated agricultural pesticide mass applied (EPest-high method, summed 2014-2018), making it one of ID's five highest pesticide-use counties. Madison County, in eastern Idaho near Rexburg, is known for seed potato production alongside grain and dairy farming.

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides in **Madison County, Idaho** during 2018, as reported by [USGS Estimated Annual Agricultural Pesticide Use](#) data. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, and each compound's chemical class (abbreviated; see the key in Table 1 and the full classification, with a separate functional class column, in `data/pesticide_classification.csv`, shared with every chapter in this repository). Rankings and estimates are based on the *EPest-high method* described by [Thelin and Stone \(2013\)](#) and [Baker and Stone \(2015\)](#):

“EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD.” -[Pesticide National Synthesis Project webpage](#)

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. “detailed interpretation of where and how much use occurs within a county is not appropriate”).
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 -included windows once a complete 2019 release is available.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
CPX	Chlorophenoxy compound
DIP	Dipyridyl compound
DTC	Dithiocarbamates
INO	Inorganic compounds
MIC	Microbial
CB	N-methyl carbamates (AChE inhibitor)
OGM	Organo-metallic compound
OC	Organochlorine compound
OP	Organophosphorous compound
OTH	Other

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)
(continued)

Code	Chemical class
PYR	Pyrethrins/Pyrethroids
TC	Thiocarbamates
TRZ	Triazines

Table 2: Top 80 EPest-high Estimates, Madison County, ID, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
1	METAM POTASSIUM	DTC	789129	360850	252584
2	METAM	DTC	368423	543566	571822
3	SULFURIC ACID	INO	153372	286860	318499
4	CHLOROPICRIN	OC	150580	66660	50000
5	DICHLOROPROPENE	OC	81246	127767	115356
6	CHLOROTHALONIL	OTH	11150	10278	9163
7	PENDIMETHALIN	OTH	9282	9008	8156
8	METRIBUZIN	TRZ	7416	7706	7093
9	EPTC	TC	6633	8661	8966
10	2,4-D	CPX	6477	5161	4716
11	GLYPHOSATE	OTH	6425	6089	6860
12	MCPA	CPX	6268	4644	6338
13	MALEIC HYDRAZIDE	OTH	5310	3475	3937
14	DIMETHENAMID & DIMETHENAMID-P	OTH	4329	4716	3934
15	DIMETHENAMID-P	OTH	4329	4575	3849
16	PHOSPHORIC ACID	INO	4174	4063	6694
17	METHOMYL	CB	3882	3258	2520
18	FLUROXYPYR	OTH	3767	3012	2888
19	METOLACHLOR & METOLACHLOR-S	OTH	3766	2942	2594
20	MANCOZEB	DTC	3660	3893	7323
21	METOLACHLOR-S	OTH	3406	2739	2371
22	BROMOXYNIL	OTH	3109	2561	3231
23	QUINTOZENE	OTH	2104	1170	1170
24	PYRIMETHANIL	OTH	1960	1668	1614
25	DIQUAT	DIP	1747	1672	2038
26	PROPARGITE	OTH	1492	1492	2133
27	GLUFOSINATE	OTH	1428	714	557
28	BOSCALID	OTH	1417	1500	1542
29	AZOXYSTROBIN	OTH	1296	1088	1181
30	PINOXADEN	OTH	1285	1146	1062
31	IMIDACLOPRID	OTH	1162	1149	1342
32	FLUOPYRAM	OTH	932	774	642
33	PYRACLOSTROBIN	OTH	874	717	728
34	DICAMBA	OTH	783	404	321
35	CHLORPYRIFOS	OP	768	690	635
36	OXAMYL	CB	762	726	2609
37	ACETOCHLOR	OTH	720	298	289
38	TRIFLURALIN	OTH	652	729	527
39	ATRAZINE	TRZ	521	670	413
40	PROPICONAZOLE	OTH	479	407	465
41	FLUTOLANIL	OTH	473	1199	1183
42	CLOPYRALID	OTH	391	298	355
43	PARAQUAT	DIP	362	1111	1434
44	METOLACHLOR	OTH	360	203	222
45	MEFENOXAM	OTH	353	510	477
46	CARFENTRAZONE-ETHYL	OTH	341	238	162
47	METIRAM	DTC	330	330	575

Table 2: Top 80 EPest-high Estimates, Madison County, ID, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) (*continued*)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
48	CYFLUTHRIN	PYR	329	157	97
49	FLUXAPYROXAD	OTH	317	222	167
50	HEXAZINONE	TRZ	266	294	194
51	TRI-ALLATE	TC	264	206	251
52	DIMETHOATE	OP	257	1669	2070
53	FLUAZINAM	OTH	257	258	432
54	DIFENOCONAZOLE	OTH	241	190	287
55	PROTHIOCONAZOLE	OTH	223	151	118
56	TEBUCONAZOLE	OTH	223	137	87
57	DIURON	OTH	222	285	245
58	METALAXYL	OTH	216	216	477
59	THIFENSULFURON	OTH	206	217	168
60	MANDIPROPAMID	OTH	199	158	202
61	CYPROCONAZOLE	OTH	187	111	111
62	SPIROTETRAMAT	OTH	171	143	88
63	COPPER HYDROXIDE	INO	126	410	441
64	PICOXYSTROBIN	OTH	120	83	125
65	CYHALOTHRIN-LAMBDA	PYR	118	203	146
66	LINURON	OTH	115	123	76
67	TRIBENURON METHYL	OTH	112	106	91
68	HYDROGEN PEROXIDE	INO	104	104	104
69	PYRASULFOTOLE	OTH	102	71	64
70	FLUCARBAZONE	OTH	94	55	33
71	ZETA-CYPERMETHRIN	PYR	92	58	56
72	THIOPHANATE-METHYL	OTH	90	341	227
73	PICLORAM	OTH	89	34	21
74	CLOTHIANIDIN	OTH	76	76	42
75	ETHALFLURALIN	OTH	74	333	254
76	CLETHODIM	OTH	69	43	57
77	PETROLEUM OIL	OTH	66	36713	22046
78	ZOXAMIDE	OTH	65	210	307
79	RIMSULFURON	OTH	64	60	84
80	HEXYTHIAZOX	OTH	62	34	30

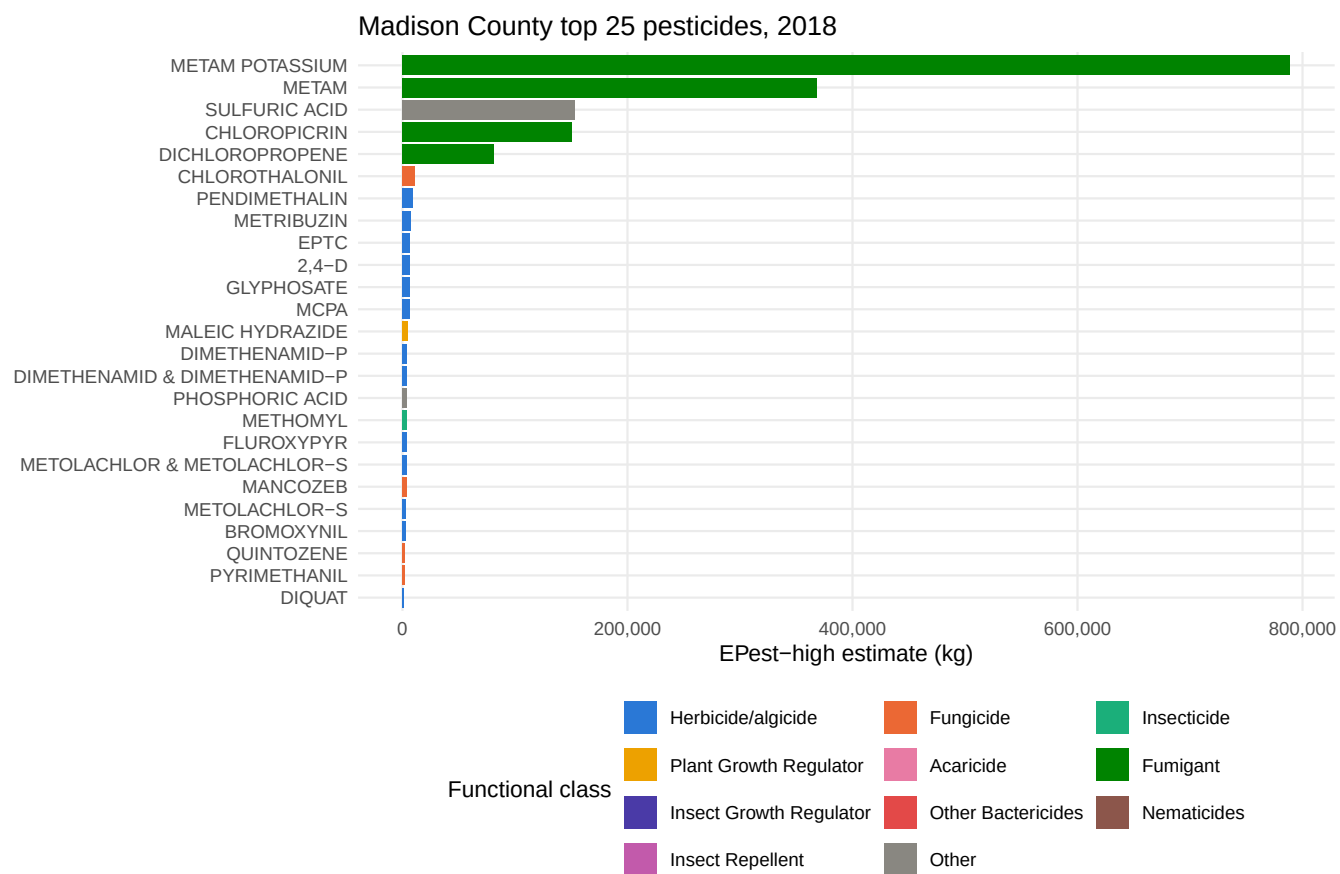


Figure 1: Top 25 compounds in Madison County by E Pest-high mass applied, most recent year, colored by pesticide functional class.

Appendices

Table 3: Top 50 Madison County Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	METAM POTASSIUM	789129	789129
2	METAM	368423	368423
3	SULFURIC ACID	153372	153372
4	CHLOROPICRIN	17	150580
5	DICHLOROPROPENE	81246	81246
6	CHLOROTHALONIL	11150	11150
7	PENDIMETHALIN	8733	9282
8	METRIBUZIN	7382	7416
9	EPTC	6633	6633
10	2,4-D	6434	6477
11	GLYPHOSATE	5566	6425
12	MCPA	6268	6268
13	MALEIC HYDRAZIDE	0	5310
14	DIMETHENAMID & DIMETHENAMID-P	4254	4329
15	DIMETHENAMID-P	4254	4329
16	PHOSPHORIC ACID	4174	4174
17	METHOMYL	0	3882
18	FLUROXYPYR	3761	3767
19	METOLACHLOR & METOLACHLOR-S	3536	3766
20	MANCOZEB	3660	3660
21	METOLACHLOR-S	3220	3406
22	BROMOXYNIL	3109	3109
23	QUINTOZENE	0	2104
24	PYRIMETHANIL	1960	1960
25	DIQUAT	1747	1747
26	PROPARGITE	0	1492
27	GLUFOSINATE	1428	1428
28	BOSCALID	1417	1417
29	AZOXYSTROBIN	1296	1296
30	PINOXADEN	1285	1285
31	IMIDACLOPRID	1162	1162
32	FLUOPYRAM	932	932
33	PYRACLOSTROBIN	871	874
34	DICAMBA	66	783
35	CHLORPYRIFOS	759	768
36	OXAMYL	762	762
37	ACETOCHLOR	0	720
38	TRIFLURALIN	66	652
39	ATRAZINE	0	521
40	PROPICONAZOLE	478	479
41	FLUTOLANIL	473	473
42	CLOPYRALID	371	391
43	PARAQUAT	2	362
44	METOLACHLOR	317	360
45	MEFENOXAM	353	353
46	CARFENTRAZONE-ETHYL	0	341
47	METIRAM	0	330
48	CYFLUTHRIN	0	329
49	FLUXAPYROXAD	316	317
50	HEXAZINONE	0	266

Table 4: Top 50 Madison County Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	METAM	543566	543566
2	METAM POTASSIUM	360850	360850
3	SULFURIC ACID	286860	286860
4	DICHLOROPROPENE	127767	127767
5	CHLOROPICRIN	7933	66660
6	PETROLEUM OIL	447	36713
7	ETHOPROPHOS	0	24880
8	PROPAMOCARB HCL	0	21784
9	CHLOROTHALONIL	10278	10278
10	PENDIMETHALIN	8218	9008
11	EPTC	8536	8661
12	METRIBUZIN	7683	7706
13	GLYPHOSATE	5787	6089
14	2,4-D	4753	5161
15	DIMETHENAMID & DIMETHENAMID-P	4506	4716
16	MCPA	4639	4644
17	DIMETHENAMID-P	4506	4575
18	PHOSPHORIC ACID	4063	4063
19	MANCOZEB	3893	3893
20	MALEIC HYDRAZIDE	0	3475
21	METHOMYL	1	3258
22	FLUROXYPYR	3008	3012
23	METOLACHLOR & METOLACHLOR-S	2710	2942
24	METOLACHLOR-S	2593	2739
25	BROMOXYNIL	2558	2561
26	DIQUAT	1672	1672
27	DIMETHOATE	290	1669
28	PYRIMETHANIL	1668	1668
29	BOSCALID	1500	1500
30	PROPARGITE	0	1492
31	FLUTOLANIL	1199	1199
32	QUINTOZENE	118	1170
33	IMIDACLOPRID	1149	1149
34	PINOXADEN	1146	1146
35	PARAQUAT	8	1111
36	AZOXYSTROBIN	1088	1088
37	SPIROMESIFEN	0	833
38	FLUOPYRAM	774	774
39	TRIFLURALIN	140	729
40	OXAMYL	726	726
41	PYRACLOSTROBIN	714	717
42	GLUFOSINATE	714	714
43	CHLORPYRIFOS	500	690
44	ATRAZINE	11	670
45	CYAZOFAMID	0	652
46	MEFENOXAM	510	510
47	NALED	0	440
48	DIMETHENAMID	0	424
49	COPPER HYDROXIDE	389	410
50	PROPICONAZOLE	406	407

Table 5: Top 50 Madison County Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	METAM	571822	571822
2	SULFURIC ACID	318499	318499
3	METAM POTASSIUM	252584	252584
4	DICHLOROPROPENE	115356	115356
5	CHLOROPICRIN	5955	50000
6	PETROLEUM OIL	283	22046
7	PROPAMOCARB HCL	0	21784
8	ETHOPROPHOS	0	16909
9	CHLOROTHALONIL	9163	9163
10	EPTC	8777	8966
11	PENDIMETHALIN	7613	8156
12	MANCOZEB	7323	7323
13	METRIBUZIN	7070	7093
14	GLYPHOSATE	6673	6860
15	PHOSPHORIC ACID	6694	6694
16	MCPA	6331	6338
17	2,4-D	4454	4716
18	MALEIC HYDRAZIDE	18	3937
19	DIMETHENAMID & DIMETHENAMID-P	3791	3934
20	DIMETHENAMID-P	3791	3849
21	BROMOXYNIL	3189	3231
22	FLUROXYPYR	2886	2888
23	OXAMYL	2609	2609
24	METOLACHLOR & METOLACHLOR-S	2294	2594
25	METHOMYL	2	2520
26	METOLACHLOR-S	2214	2371
27	PROPARGITE	0	2133
28	DIMETHOATE	177	2070
29	DIQUAT	2038	2038
30	PYRIMETHANIL	1614	1614
31	BOSCALID	1542	1542
32	PARAQUAT	14	1434
33	IMIDACLOPRID	1342	1342
34	FLUTOLANIL	1183	1183
35	AZOXYSTROBIN	1181	1181
36	QUINTOZENE	118	1170
37	PINOXADEN	1060	1062
38	SODIUM CHLORATE	848	848
39	PYRACLOSTROBIN	723	728
40	FLUOPYRAM	642	642
41	CHLORPYRIFOS	445	635
42	METIRAM	465	575
43	GLUFOSINATE	557	557
44	TRIFLURALIN	113	527
45	SPIROMESIFEN	0	490
46	METALAXYL	2	477
47	MEFENOXAM	477	477
48	PROPICONAZOLE	464	465
49	COPPER HYDROXIDE	428	441
50	TERBACIL	0	441